STPS: Spatio-Temporal Proactive Scheduling for Spiking Neural Networks on Compute-in-Memory Clusters

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Abstract

Spiking Neural Networks (SNNs) deployed on Compute-in-Memory (CIM) clusters present a scheduling problem that does not appear in conventional DNN serving: each model's network-on-chip (NoC) traffic is both *input-dependent* and *time-varying*, so a tenant that looks idle on average can saturate a card's NoC during a microsecond-scale spike burst. A formulation that schedules every spike is a time-dependent mixed-integer non-linear program (MINLP) and is intractable at admission time; reactive runtime mitigation only observes congestion once the network has already saturated.

This paper makes three contributions. First, it introduces a compact SNN workload fingerprint: an offline profiler reduces each calibration trace to a traffic timeline $E^{(t)}$, burstiness β, mean active components $\overline{K}$, and a recorded hotspot indicator, so admission decisions no longer inspect the full spike trace. Second, it presents **STPS**, a two-stage admission scheduler that uses the fingerprint to choose a card and then search a bounded start offset that places the task's traffic into a valley of the selected card's forecast. Third, it evaluates STPS against four fingerprint-blind baselines across Poisson, bursty, and mixed arrivals at 4-card and 16-card scales. Congestion is where STPS wins: it is the only policy that lowers NoC congestion — on both the congestion ratio and the queue wait — in all six arrival/scale cells, cutting the congestion ratio by 7–11 % and mean queue wait by 8–18 % relative to the best baseline in each cell. It buys this relief almost for free: throughput stays within 0.3 %, and at 16 cards STPS additionally attains the lowest across-card CV and highest fairness of any policy, so the win does not trade off load balance. The whole per-card congestion distribution shifts down, not just its mean. STPS pays only in the tail, adding 2–3 ticks at $p99$ on the worst $\sim$1 % of tasks.

1 Introduction

Energy-efficient deep learning has rekindled interest in Spiking Neural Networks (SNNs) on neuromorphic Compute-in-Memory hardware. Production-class platforms now span academic and industrial designs — TrueNorth [10], Loihi and Loihi-2 [1, 13], SpiNNaker-2 [9], and Darwin3 [8] — and their event-driven execution model converts a model's idle neurons into idle silicon. The promise is attractive enough that a *Neuromorphic Computing-as-a-Service* cluster — many tenant SNNs sharing one bank of CIM cards — is a plausible deployment target for the next generation of neuromorphic systems.

Scheduling on such a cluster looks superficially like a DNN serving problem and is fundamentally different. A DNN model has a deterministic compute graph: at admission, a placement decision based on FLOPs, memory, and bandwidth is correct for the entire run, and predictable execution is the design goal of recent serving stacks such as Clockwork [5]. An SNN's per-tick activity, by contrast, is a function of the input. A vision classifier on a quiescent frame fires a handful of spikes; the same model on a high-contrast frame can inject orders of magnitude more traffic into the NoC. Even within one inference, spikes arrive in narrow temporal bursts driven by neural dynamics rather than uniformly across the inference window. Two consequences follow. First, capacity-based placement — best-fit on free cores, dominant-resource fairness on weighted resources [3], power-of-two-choices on instantaneous load [11] — spreads cards evenly by core count yet leaves them temporally hot: the bursts of co-located tenants align in time and the per-card NoC saturates even though average occupancy looks balanced. A scheduler blind to the time dimension cannot see this collision coming. Second, treating each spike as a scheduling unit is infeasible: at admission the scheduler does not yet know which spikes will fire, and even if it did the resulting MINLP would not solve in the per-task latency budget of a serving system.

A pragmatic scheduler must therefore reason about *distributions* of spike activity, not individual spikes. The information that distinguishes a "quiet" task from a bursty one is statistical, available before the task arrives, and small enough to consult in constant time, independent of the trace length. We make this concrete: a Discrete-Time Dynamic Graph (DTDG) view of an SNN trace yields four compact fingerprints — the traffic

timeline $E^{(t)}$, global burstiness β, mean active components $\overline{K}$, and a per-population hotspot indicator — of which the first three suffice to drive admission-time placement and timing decisions without re-examining the underlying weight tensor (the hotspot indicator is recorded for a future spatial-splitting extension).

The resulting scheduler, **STPS** (Spatio-Temporal Proactive Scheduling), takes these fingerprints as input. A macro-dispatch stage scores each card with a closed-form weighted sum of fragmentation and burstiness pressure, picking the card whose fragmentation matches the task's topology and whose forecast burst-density tolerates the task's β. A phase-shift stage then searches at most $D_{\max}$ start offsets and commits the offset whose addition to the card's forecast minimises the peak ($O(D_{\max} \cdot H)$). The two stages address orthogonal failure modes — spatial fragmentation and temporal collision — and are the minimum mechanism that closes the gap between capacity-based DNN scheduling and SNN execution dynamics.

Contributions.

- A *workload fingerprint* that distils an SNN's spatio-temporal injection profile into four scalars/vectors derivable from a calibration trace, sufficient to support admission-time decisions whose cost is constant in the model size and independent of the DTDG length (Section 5.2).
- A two-stage online scheduler that consumes the fingerprint to dispatch tasks across cards and to shift their start offsets, with both stages reducing to closed-form scoring or a bounded search over $D_{\max}$ offsets (Sections 5.3.1 and 5.3.2).
- An evaluation on real and synthetic SNN traces showing that STPS lowers NoC congestion by 7–11 % and queue wait by 8–18 % across arrival modes — the only baseline-competitive scheduler to do so in every cell — at under 0.3 % throughput cost and, at scale, with a load balance no worse (in fact better) than the capacity baselines, while isolating each stage's contribution, quantifying scheduling overhead and robustness to fingerprint drift, and characterising the long-tail delay cost the scheduler pays for these gains (Section 7).

The rest of the paper is organised as follows. Section 2 explains why both capacity-based and per-spike scheduling fail on SNN/CIM clusters; Section 3 positions STPS against prior work; Section 4 fixes notation; Section 5 presents the offline fingerprint and the two online stages; Section 6 describes the prototype; Section 7 reports the RQ1–RQ4 results and the long-tail trade; Section 8 concludes.

2 Background and Motivation

This section establishes what makes SNN serving on CIM clusters distinct from DNN serving, and why two natural design points — capacity-based scheduling and per-spike optimisation — fail. The argument frames the design constraint that drives Section 5: the scheduler needs a workload abstraction that is statistical, small, and consultable in $O(1)$.

2.1 SNN Execution on CIM Clusters

A CIM card hosts a 2D mesh of cores; each core stores a fixed-size slice of synaptic weights in-memory and routes spikes to its neighbours over an on-chip network. This memory-co-located organisation is shared across the production neuromorphic substrates we target — Loihi/Loihi-2 [1, 13], TrueNorth [10], SpiNNaker-2 [9], and Darwin3 [8] — and dictates three properties that shape the scheduling problem.

(i) Bound state. A neural *population* is mapped onto one or more cores at compile time. When any pre-synaptic neuron of population j fires, the post-synaptic membrane updates run locally and out-going spikes are injected into the NoC towards population j's downstream targets. Once the populations of a task are pinned, the task cannot migrate mid-inference without a costly state copy [17], so admission-time decisions are largely irrevocable.

(ii) NoC-dominated contention. The cluster aggregates M such cards; an inter-card interconnect is treated as a separate bandwidth pool. The dominant shared resource is the per-card NoC: when instantaneous injection exceeds its bandwidth ceiling, downstream spikes are queued and tail latency grows.

(iii) Capacity is the wrong yardstick. A card's free-core count is a poor proxy for its real load, since spike injection is sparse and time-varying. Two cards with identical free-core counts can differ by an order of magnitude in their actual NoC pressure once their resident tasks burst together.

2.2 Why Capacity-Based Scheduling Fails

Schedulers from the DNN serving literature place tasks based on scalar resource demands: best-fit or first-fit on heterogeneous resource vectors [14], dominant-resource fairness on a weighted vector [3], power-of-two-choices on instantaneous load [11]. All three are blind to the time dimension. Two tasks that look identical to DRF can have completely different traffic shapes; if the scheduler co-locates a steady-flat tenant and a bursty tenant whose peaks happen to align, the card's NoC saturates even though its mean utilisation is well below capacity. We measure this directly in Section 7.2: under bursty arrivals on a capacity-balanced 16-card cluster the busiest card's NoC injection exceeds the workload's own p75 cap in ∼98 % of steady-state ticks, and runs 3–5× the cluster mean even at mean utilisations as low as 46 %. Migration-based fixes proposed for GPU clusters [16, 17] can rebalance after

the fact, but on CIM hardware the population state is large and the migration cost dwarfs the window in which the move would have helped.

2.3 Why Schedule-Every-Spike Fails

The opposite extreme — treating each spike as a scheduling decision — is computationally infeasible. Even at modest activity levels (a thousand spikes per task per tick, hundreds of co-resident tasks), a per-spike formulation produces a time-dependent MINLP whose state space exceeds the per-task admission budget by several orders of magnitude. Equally fatal, the scheduler does not know which spikes will fire at admission: the firing pattern depends on the input. A heuristic that schedules at this granularity must either delay admission until the firing trace is observed (eliminating any chance of proactive placement) or guess (eliminating the precision that motivated the per-spike view in the first place).

2.4 The Missing Middle

The scheduler we need lives between these extremes: it must reason about each task's spike activity as a *distribution* that is known at admission time, small enough to evaluate in $O(1)$, and rich enough to expose both spatial topology (which cards' fragmentation matches the task) and temporal shape (where in the card's forecast a new task fits). The next two sections describe how STPS realises this middle: Section 5.2 reduces an SNN's DTDG to four such quantities, and Sections 5.3.1–5.3.2 consume them in closed form.

3 Related Work

Cluster scheduling for DNN serving. Most production cluster schedulers approach placement as bin-packing under multi-dimensional capacity constraints. Best-fit and first-fit on heterogeneous resource vectors [7, 14], dominant-resource fairness [3], and multi-resource generalisations [4] pick a card whose free vector dominates the task's request. These policies are blind to temporal load shape and constitute our baselines in Section 7. Schedulers specialised for DNN training (Gandiva [17], Synergy [11], heterogeneity-aware placement [12]) exploit job-level signals — iteration time, GPU sharing, accelerator heterogeneity — but their unit of work is a deterministic GPU kernel whose footprint does not vary with the input. Predictable DNN serving systems such as Clockwork [5] go further still and make execution time itself a constant by centralising scheduling decisions; this is feasible because the underlying kernel cost is data-independent. None of these systems surfaces a per-tick traffic profile to the scheduler, which is the abstraction STPS relies on, because their workload model does not have one.

Cluster-level admission control. A complementary line of work tackles bursty *arrivals* by predicting future load and provisioning ahead. Workload prediction in serving systems [16] estimates request-rate spikes and pre-scales replicas; admission-control systems for cloud quotas [15] reason about queue depths and fairness across tenants. Both predict the *number* of jobs, not the temporal shape of any single job's hardware demand. The phase-shift stage of STPS is closer to packing in the temporal dimension: rather than predicting future arrivals, it consumes a per-task forecast and chooses a start offset that avoids collision with the card's existing schedule. The closest classical analogue is gang scheduling on parallel machines, but gang scheduling coordinates start times *across cards* to enable communication; STPS coordinates start times *on a single card* to spread NoC pressure.

Neuromorphic mapping and validation. Neuromorphic compilers for production substrates (TrueNorth [10], Loihi [1, 13], SpiNNaker-2 [9], Darwin3 [8]) focus on a static problem: how to partition and place a single model's populations onto cores so that on-chip routing meets latency and fan-out constraints. Validation work on SpiNNaker [7] characterises spike-traffic distributions at this single-model granularity. These tools answer "how does one model fit one chip", which is orthogonal to the multi-tenant placement and timing decisions STPS makes at admission. We treat compiled per-model placements as inputs — the per-population centrality and connected-component structure extracted from the DTDG presupposes that this static mapping has already happened — and ask the next-level question of how to dispatch many such pre-compiled models across many cards.

Fingerprint-driven scheduling. Compressing a workload to a small set of summary statistics is a recurring idea in cluster scheduling and load balancing. STPS's contribution is the choice of fingerprint: four quantities that together capture both the spatial topology ($\overline{K}$, $\mathbf{c}^*_{\max}$) and temporal shape ($E^{(t)}$, β) of an SNN's spike traffic, in a representation small enough to drive admission-time decisions and physical enough to derive from a calibration trace — specifically, from SpikingJelly [2] runs of architectures such as Spikformer [18] — without bespoke instrumentation.

4 Preliminaries

This section fixes the notation used by the rest of the paper. Symbols defined here recur in Section 5 (system) and Section 7 (evaluation); we do not re-introduce them later.

4.1 Workload Model

A submitted task τ is a pre-compiled SNN already partitioned by the vendor toolchain into a set $\mathcal{V}_\tau$ of hardware-aligned pop-

ulations, each population fitting in one CIM core. An *inference* runs for T discrete ticks. At each tick t, only a subset of populations fires; the resulting per-tick weighted graph $G_\tau^{(t)} = (\mathcal{V}_\tau, \mathcal{E}_\tau^{(t)}, W_\tau^{(t)})$ records the active inter-population edges and the expected spike traffic $W_\tau^{(t)}[i,j]$ flowing from population i to population j. The sequence $\mathcal{G}_\tau = \{G_\tau^{(1)}, \ldots, G_\tau^{(T)}\}$ is a Discrete-Time Dynamic Graph (DTDG); expectations are taken over a calibration set drawn from the model's evaluation distribution, so $\mathcal{G}_\tau$ depends on the model, not on any specific input.

We never use $\mathcal{G}_\tau$ directly at scheduling time. Instead, Section 5.2 reduces it to four *fingerprint* quantities: the traffic timeline $E_\tau^{(t)} \in \mathbb{R}^T$, global burstiness β_τ, mean active components $\overline{K}_\tau$, and per-population hotspot indicator $\mathbf{c}^*_{\max,\tau} \in \mathbb{R}^{|\mathcal{V}_\tau|}$. The scheduler reads only these.

4.2 Cluster Model

The cluster is a set of M homogeneous cards indexed by $m \in \{1, \ldots, M\}$. Each card holds C CIM cores arranged in a 2D mesh and a NoC with per-card injection ceiling $BW_{\max}$ (spikes/tick). For card m we track its free-core occupancy and, derived from it, the largest contiguous free-block ratio $\rho_m \in [0,1]$ used by Stage 1 dispatch. The card also maintains a per-tick injection forecast $E_m^{(t)} \in \mathbb{R}^H$ over a sliding horizon H, accumulated from all currently-scheduled tasks' offsetted timelines. When the instantaneous demand on card m exceeds $BW_{\max}$, the excess is queued in a pending-traffic buffer rather than dropped; the engine drains it at the cap rate.

Inter-card transfers are modelled as a separate resource pool but never used by our scheduler, which enforces single-card placement: each task lives entirely on one card for the duration of its inference.

4.3 Scheduling Problem

Tasks arrive online according to an arrival process. On each arrival of τ, the scheduler must (a) commit a card $m^*(\tau) \in \{1, \ldots, M\}$ for placement and (b) commit a start offset $\Delta t^*(\tau) \in \{0, 1, \ldots, D_{\max}\}$ after which τ's timeline begins injecting into $E_{m^*}^{(t)}$. Both decisions are irrevocable for the duration of the inference: re-placement requires copying the population state and is prohibitive on CIM hardware.

A scheduler is admissible if, for every τ, it returns $(m^*, \Delta t^*)$ within a per-task budget that is constant in the model size and independent of the DTDG length: it may scan the M cards and a bounded offset window, but never re-reads the underlying weight tensor or spike trace. The system-level objective is to minimise NoC contention — specifically, the fraction of offered NoC work left queued behind $BW_{\max}$ and the mean wait in the pending queue — subject to maintaining cluster throughput and without degrading across-card load balance. Section 5 describes the policy STPS uses to make these two decisions; Section 7 measures it against the objectives above.

5 System Design

5.1 Overview

Scheduling SNN inference on a multi-card CIM cluster is hard because spike traffic is both *input-dependent* and *time-varying*: the same model exhibits different injection patterns across inputs and across ticks within one inference. A formulation that schedules every spike exactly is a time-dependent MINLP and is intractable at admission time; reactive runtime mitigation observes congestion only after the NoC has already saturated. STPS resolves this tension by separating what can be known offline from what must be decided online.

Framework. Figure 1 shows the two-phase pipeline. *Offline*, a profiler runs once per model: it traces the SNN over a calibration set, lifts the trace into a Discrete-Time Dynamic Graph (DTDG), and reduces the DTDG to a four-quantity *fingerprint* stored as a small memory-mapped artefact. *Online*, on each task arrival the scheduler reads only the fingerprint and runs two stages in sequence: macro-card dispatch picks a card m^*, then phase-shift picks a start offset Δt^*. The pair $(m^*, \Delta t^*)$ is committed to the chosen card's forecast and returned to the runtime. Stage 1 scores the M candidate cards in $O(M)$ and Stage 2 scans the $D_{\max}+1$ offsets over the H-tick horizon in $O(D_{\max} \cdot H)$; both are constant in the model size and independent of the DTDG length, so admission cost does not grow with model complexity.

Stage 1 — Macro-Card Dispatch. The first stage chooses *where* the task lands. For each card m with sufficient free capacity, it computes a single scalar score $\mathcal{S}_m(\tau)$ that combines two pressures: (i) how well the task's topological cohesion $\overline{K}_\tau$ matches the card's largest contiguous free-block ratio ρ_m, and (ii) whether the card already hosts bursty workloads when τ is itself bursty ($\beta_\tau > \beta_{\text{hi}}$). The card minimising $\mathcal{S}_m$ wins. The score is a weighted sum rather than a lexicographic rule, so it degrades smoothly between empty and saturated regimes without policy switching.

Stage 2 — Temporal Phase-Shifting. The second stage chooses *when* the task starts. Spatial dispatch alone does not prevent two bursty tasks from peaking on the same tick; Stage 2 exploits the fact that the task's traffic timeline $E_\tau^{(t)}$ is known at admission. It scans the $D_{\max}+1$ candidate offsets $\Delta t \in \{0, \ldots, D_{\max}\}$ and commits the one whose superposition $E_{m^*}^{(t)} + E_\tau^{(t-\Delta t)}$ has the smallest peak under the bandwidth ceiling $BW_{\max}$. The chosen offset is added back to $E_{m^*}^{(t)}$, updating the forecast that the next admission will see.

The rest of this section justifies the fingerprint design (Section 5.2) and details each online stage (Sections 5.3.1 and 5.3.2); the hotspot indicator and discussion follow in Sections 5.4 and 5.5.

5.2 Workload Fingerprint

We model an SNN inference as a Discrete-Time Dynamic Graph (DTDG) $\mathcal{G} = \{G^{(1)}, \ldots, G^{(T)}\}$, where each snapshot $G^{(t)} = (\mathcal{V}, \mathcal{E}^{(t)}, W^{(t)})$ shares the same vertex set $\mathcal{V}$ of hardware-aligned neural populations (each population fits within one CIM core) and a per-tick weight tensor $W_{ij}^{(t)}$ measuring the expected spike traffic on edge $i \to j$. The expectation is taken over a calibration set drawn from the model's evaluation distribution, so the fingerprint is independent of any single input.

From this trace we extract three quantities that the online scheduler consumes:

- **Traffic timeline $E^{(t)} \in \mathbb{R}^T$.** The total spike traffic at tick t, averaged over the calibration set. $E^{(t)}$ is the only fingerprint that retains temporal shape.
- **Global burstiness $\beta = \max_t E^{(t)} / \overline{E}$.** Peak-to-mean ratio of the traffic timeline; $\beta \approx 1$ for steady workloads, $\beta \gg 1$ for event-driven bursts.
- **Mean active components $\overline{K}$.** Mean number of connected components in the active-edge subgraph of $G^{(t)}$, averaged over t. $\overline{K} \to 1$ indicates a topologically cohesive model; $\overline{K} \gg 1$ indicates several concurrent, weakly-coupled subflows.

A fourth quantity, the per-population maximum in-eigenvector centrality $\mathbf{c}^*_{\max}[i] = \max_t c_i^{(t)}$, is recorded as a hotspot indicator and discussed in Section 5.4.

The fingerprint is on the order of tens of kilobytes per model and is computed once at deployment time; subsequent scheduling decisions read it as a memory-mapped artefact.

5.3 Online Scheduler

This subsection details the two stages summarised in Section 5.1.

5.3.1 Stage 1: Macro-Card Dispatch

The cluster constraint we enforce is single-card placement: each model resides entirely on one card. The dispatch problem therefore reduces to selecting one card among those with sufficient free cores and memory for τ. We score each feasible card m by

$$\mathcal{S}_m(\tau) \;=\; \alpha \left| \rho_m - \tfrac{1}{\overline{K}_\tau} \right| \;+\; \gamma \beta_m \cdot \mathbb{1}[\beta_\tau > \beta_{\text{hi}}], \qquad (1)$$

Algorithm 1 Phase-Shift Selection

Require: Background E_m, task E_τ, budget $D_{\max}$, ceiling $BW_{\max}$
Ensure: Offset $\Delta t^* \in [0, D_{\max}]$ (offset 0 if none fits under $BW_{\max}$)

```
p* ← ∞, Δt* ← 0                          ▷ default: admit unshifted
for Δt = 0 to D_max do
    p ← max_t (E_m^(t) + Shift(E_τ, Δt)^(t))
    if p ≤ BW_max and p < p* then
        p* ← p, Δt* ← Δt
    end if
end for
E_m ← E_m + Shift(E_τ, Δt*)
return Δt*
```

where ρ_m is the largest contiguous free-block ratio on card m and β_m is its running mean burstiness. The first term matches the task's topological cohesion against the card's available shape: cohesive tasks ($\overline{K}_\tau \to 1$) are drawn to cards with large contiguous blocks ($\rho_m \to 1$), while decoupled tasks ($\overline{K}_\tau \gg 1$) absorb fragmented cards without penalty. The second term keeps high-burstiness tasks away from cards that already host bursty workloads; it activates only above a threshold β_{hi} so that flat traffic incurs no temporal penalty. The card with minimum $\mathcal{S}_m$ wins.

5.3.2 Stage 2: Temporal Phase-Shifting

Cards co-locate multiple tasks, and their traffic timelines superpose on the same on-chip network. Spatial dispatch alone does not prevent two bursty tasks from peaking on the same tick. Stage 2 exploits the fact that the timeline $E_\tau^{(t)}$ is known at admission to delay τ's injection so that its peaks fall into the valleys of the existing card-level forecast.

Let $E_m^{(t)}$ be the forecasted background traffic on the selected card m and $D_{\max}$ the deadline-bounded shift budget. The optimal offset is

$$\Delta t^* \;=\; \arg \min_{\Delta t \in [0, D_{\max}]} \; \max_t \left(E_m^{(t)} + E_\tau^{(t-\Delta t)} \right), \qquad (2)$$

subject to the resulting peak staying below the per-card bandwidth ceiling $BW_{\max}$. When no offset in $[0, D_{\max}]$ satisfies the ceiling, the task is admitted unshifted (offset 0) — it is never dropped — and simply contributes its share to congestion; we call this an *offset-infeasible* admission, and it is rare except at extreme load. Algorithm 1 expresses this as a single linear scan over $D_{\max}+1$ candidate offsets; the cost is $O(D_{\max} \cdot H)$ per task.

5.4 Spatial Mapping and Hotspot Indicator

Once a card and offset are fixed, populations are mapped to PIM cores by the vendor compiler under standard hardware

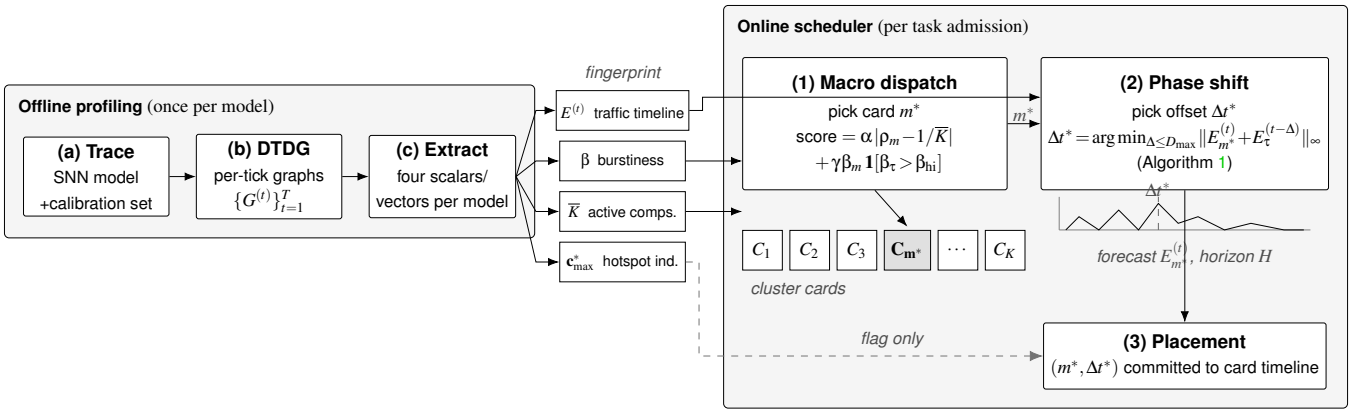


Figure 1: STPS pipeline. *Offline:* a model is traced, lifted into a discrete-time dynamic graph, and reduced to four fingerprints. *Online:* each admission runs in two stages — macro-dispatch picks a card m^* from $\{C_1, \ldots, C_K\}$ using $\overline{K}$ and β, then phase-shift picks a start offset $\Delta t^* \le D_{\max}$ that fits the task's traffic timeline into a valley of the card's forecast. The hotspot indicator $\mathbf{c}^*_{\max}$ is recorded but does not drive placement in the current system (dashed arrow).

constraints. The fingerprint exposes $\mathbf{c}^*_{\max}$ as a per-population indicator of compute concentration: a population whose centrality exceeds a threshold θ_c is a candidate for splitting across adjacent cores, trading a small spatial footprint for reduced peak SOP pressure. In the current system this flag is recorded on the task but does not drive placement; we leave the centrality-driven splitting pass, and its end-to-end evaluation, to future work.

5.5 Discussion

Two design choices warrant comment. First, all four fingerprint quantities are scalars or short vectors; the scheduler never inspects the underlying $T \times |\mathcal{V}|^2$ trace at runtime. This keeps admission cost independent of model size and is what allows STPS to interleave with arrival rates of thousands of tasks per second. Second, the macro-dispatch score in Equation 1 is deliberately a weighted sum rather than a lexicographic policy: the same scalar handles both empty and saturated regimes, so STPS degrades gracefully as cluster load increases rather than switching between policies.

6 Implementation

We implement STPS as a Python prototype on top of a discrete-event simulator of a multi-card CIM cluster. The system is partitioned into three packages so that the offline profiler, the online scheduler, and the simulation harness can evolve independently. The fingerprint extractor (FINGERPRINT/, ∼2 kLoC) lifts an SpikingJelly [2] run into a DTDG and reduces it to the four quantities of Section 5.2; the resulting artefact is persisted as a memory-mapped `.npz` blob (∼tens of kilobytes per model) and re-loaded lazily at admission time, so cold-start cost is a single file read. The online scheduler (SCHEDULE/, ∼0.8 kLoC) implements both stages and a pluggable registry that exposes RR, BestFit, DRF, P2C, and the STPS variants behind a uniform interface; the registry also hosts the stage ablations as named entry points so the evaluation harness can sweep policies without touching the simulator core.

The simulation harness (SIMULATION/, ∼0.5 kLoC) implements the two-tier tick loop — per-card placement, NoC bandwidth accounting, pending-traffic queue — in pure NumPy; we deliberately avoid a hardware-cycle-accurate model because the scheduling decision is made entirely at admission, and the per-tick simulation only has to be accurate *enough* to faithfully expose the bandwidth ceiling and queue dynamics. The whole system is around 4.6 kLoC of Python and 1.8 kLoC of pytest tests, exercised through a `Makefile` that drives both the per-policy runs and the cross-scheduler comparison sweeps used in Section 7. The fingerprint extractor can also be invoked standalone (`python -m fingerprint.cli`) to produce synthetic fingerprints, which is what the steady-flat and pulse-train workloads in Section 7.1 are built from.

7 Evaluation

The evaluation answers four questions. **RQ1** (Section 7.3): does STPS reduce end-to-end NoC congestion, and at what cost? **RQ2** (Section 7.4): do macro-dispatch and phase-shift attack congestion along complementary axes, so that neither stage alone suffices? **RQ3** (Section 7.5): in which operating regimes does the relief hold, and where does it degrade? **RQ4** (Section 7.6): where in the delay distribution does STPS pay for the relief? We then report the scheduler's own overhead (Section 7.7), its robustness to fingerprint drift (Section 7.8),

its scaling to 64 cards (Section 7.9), and the simulator's fidelity and limitations (Section 7.10). A measurement of the congestion pathology on the baselines (Section 7.2) motivates the design.

The headline finding is a consistency one. STPS is the only scheduler that lowers NoC congestion — on both the congestion ratio and the mean queue wait — in every one of the six arrival/scale cells tested, while capacity-blind baselines trade the lead from cell to cell. It cuts the congestion ratio by 7–11 % and the queue wait by 8–18 % relative to the best baseline in each cell, and does so almost for free: cluster throughput stays within 0.3 %, and at 16 cards STPS additionally posts the lowest across-card CV and the highest fairness index — so the congestion relief does not come at the expense of load balance. The whole per-card congestion distribution shifts down, not just its mean. The one cost is confined and explicit: STPS spends a 2–3 tick p99 increase on the worst $\sim$1 % of tasks (Section 7.6).

7.1 Trace-Driven Setup Exposes NoC Contention

The simulator models per-card NoC pressure. We use a discrete-event, trace-driven simulator of a multi-card CIM cluster (Section 5). Each card holds 256 cores wired by a 2D mesh NoC. The engine accounts for per-card instantaneous demand, NoC bandwidth cap $BW_{\max}$, and a pending-traffic queue that absorbs over-budget injection rather than dropping the task; the queue is drained at the cap rate so no task is ever rejected and the cluster's completion rate is always 1.0. All schedulers share the simulator and the same per-card placement primitive; they differ only in the admission-time dispatch and (for STPS) phase-shift logic.

Workloads mix synthetic extremes with real SNN traces. Each task draws a fingerprint from a *mixed* set: four synthetic templates (steady-flat, two pulse trains with periods $T = 8$ and $T = 16$, and a heavy-tail burst) plus three real SpikingJelly [2] traces (Spikformer [18]-CIFAR10, QKFormer-CIFAR10, SpikingResFormer-Tiny-ImageNet). Task arrivals follow either a Poisson process, a bursty arrival pattern that bunches admissions into onset windows, or a 1:1 mix of the two.

Trace-derived NoC injection is the modeled bottleneck. The simulator abstracts the cluster at the granularity our claims live at: per-card, per-tick NoC injection. Three properties make this sufficient to support the conclusions. First, the traffic timeline $E_\tau^{(t)}$ that drives every placement and phase-shift decision is *trace-derived*, not parametric: the three real fingerprints are extracted from actual SpikingJelly spike tensors via the offline DTDG pipeline (Section 5.2), so the burst shapes the scheduler reacts to are the ones the models actually produce. The four synthetic templates exist only to sweep extremes (perfectly flat, single-period, heavy-tail) that the real traces sample but do not isolate. Second, the contended resource we model — the NoC bandwidth cap and the pending-traffic queue draining at that cap — is the resource SNN inference on CIM actually serialises on: with weights resident in memory, inference is communication- not compute-bound, so an injection-rate model captures the first-order bottleneck while remaining agnostic to per-core compute timing that scheduling does not influence. Third, the cap itself is not a tuned knob: $BW_{\max}$ is fixed at the p75 of the per-card demand observed under an *uncapped* run, a workload-derived operating point at which every scheduler — ours included — sees congestion, rather than a value chosen to flatter STPS. We report the sensitivity of our conclusions to this choice across three cap-binding regimes in Section 7.5.

The methodology supports policy-level claims. Three boundaries follow from a simulation-first study. First, the bandwidth cap is set from the workload itself (p75 of uncapped demand) rather than calibrated against a specific silicon NoC, whose effective cap depends on routing, arbitration, and contention that an injection-rate model abstracts away; we bracket this uncertainty with the cap-binding sweep of Appendix A, whose binding, light, and non-binding regimes span the congestion levels a heavier (p50-like) or lighter (p90-like) cap would induce, and STPS's qualitative ordering holds across all three. Second, the fingerprints are extracted from the same calibration traces against which the scheduler is evaluated; a deployment trace that drifts from the calibration trace would erode Stage A's topology score — an effect RQ3 characterises along the predictability axis but does not isolate as an explicit calibration-to-deployment mismatch. Third, the numbers are throughput, congestion, and delay in a discrete-event model, not wall-clock latency on hardware; closing that gap requires a cycle-accurate or silicon NoC study we leave to future work. We therefore read our results as evidence about scheduling *policy* under a faithful first-order NoC model, not as absolute performance predictions for a specific CIM part.

The default configuration fixes a congested operating point. Unless stated otherwise the cluster has $M = 4$ cards, runs for 512 ticks, $BW_{\max} = 9 \times 10^5$ (the p75 of the per-card demand observed under an uncapped run, so all schedulers see meaningful congestion), $D_{\max} = 2$, $H = 64$. We average each configuration over 5 seeds (21, 42, 99, 123, 2024) and report ± 95 % confidence half-widths where space permits; steady-state metrics skip the first and last 64 ticks of each run. The 16-card configuration scales the task count to 3200 to keep utilisation comparable.

Baselines are fingerprint-blind spatial schedulers. Round-robin (**RR**), best-fit on free cores (**BestFit**), dominant-resource fairness (**DRF**) [3], and power-of-two-choices (**P2C**) [11]. All four are spatial-only: they ignore fingerprints and inject tasks with zero offset. For the Stage-A ablation we isolate STPS's macro-dispatch as the spatial-only variant STPS-SPATIAL; for the Stage-B ablation we additionally pair each baseline with the phase-shift module (+PS). The full STPS scheduler is the STPS-SPATIAL+PS pairing.

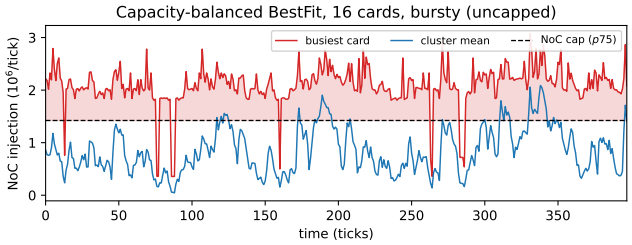


Figure 2: Per-card NoC injection over time for capacity-balanced BestFit (16 cards, bursty, uncapped). The busiest card (red) sits above the workload's own p75 bandwidth cap (dashed) in ∼98 % of steady-state ticks, while the cluster mean (blue) stays far below it: even-by-core-count packing leaves individual cards chronically saturated. The shaded area is demand the cap would have to queue.

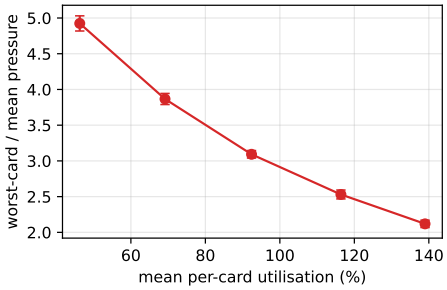


Figure 3: Worst-card-to-mean NoC pressure versus mean per-card utilisation (BestFit, 16 cards, bursty, 5 seeds). The busiest card runs 3–5× the cluster mean across the whole range, and the ratio is *highest* at low utilisation (4.9× at 46 %): the pathology is not a symptom of overload but of temporal burst alignment, which capacity balance cannot see.

Metrics lead with congestion, then throughput, tail, and balance. The metrics that carry our claims are the congestion ones: NoC *congestion ratio* (the fraction of offered NoC work left queued behind the cap) and the mean wait in the pending queue (*cong-wait*, ticks). Alongside these we report cluster *throughput* in tasks/tick and the p99 of per-task end-to-end delay, which we use to characterise the tail trade in Section 7.6. We also report the across-card load balance — coefficient of variation *CV* and Jain's fairness index *JFI* [6] (JFI $= (\sum_m L_m)^2/(M\sum_m L_m^2) \in (1/M, 1]$, 1 = perfectly balanced) — not as objectives STPS optimises but to show the congestion relief does not cost balance (at scale STPS in fact leads both). Finally, as a pure *diagnostic*, we report the max/min served-load ratio (*worst-card ratio*); it is a dispersion statistic a core-packing baseline can pin low, and STPS neither targets nor leads it. To separate Stage-B's intrinsic offset from genuine queueing, we additionally report the *cold-start-excluded throughput*, $N_{\text{completed}}/(T_{\text{steps}} - \overline{T_{\text{cold}}})$, where $\overline{T_{\text{cold}}}$ is the per-run mean start offset committed by phase-shift.

7.2 Motivation: Capacity Balance Leaves the NoC Saturated

Before evaluating STPS we quantify the problem it targets, on the schedulers it must beat. We run a capacity-balanced baseline (BestFit) at 16 cards under bursty arrivals with the bandwidth cap removed, so the recorded per-card injection is the workload's true *demand* rather than the cap-clipped served rate, and we ask a single question: does packing tasks evenly by core count keep each card's NoC below its bandwidth?

The mean hides the saturation. Figure 2 shows the answer is no. The busiest card's injection exceeds the workload's own p75 demand cap in 98 % of steady-state ticks, even though the cluster-mean injection sits at less than half the cap throughout. Averaged over cards or over time, the cluster looks comfortably provisioned; instantaneously, one card is almost always over its NoC budget. Figure 3 sharpens the point across load levels: the worst-card-to-mean ratio is 3–5× at every utilisation and is *largest* at the lowest utilisation (4.9× at 46 %), because at light load a single co-located pair of aligned bursts dominates a card while its neighbours idle. This is exactly the failure mode a time-blind capacity scheduler cannot avoid: it balances the resource it can see (cores, mean load) and is blind to the one that saturates (instantaneous NoC injection). The rest of this section shows that a fingerprint-aware, phase-shifting scheduler closes this gap — lowering the congestion that Figure 2 makes visible in every operating cell.

7.3 RQ1: STPS Consistently Reduces NoC Congestion

Congestion falls in every cell. Table 1 reports STPS against the four baselines on the same workload at two cluster sizes. The primary result is in the congestion columns: STPS wins all six arrival/scale cells on both the congestion ratio and the mean queueing wait, and no baseline wins on either metric in any cell. The margin is 7–11 % on the ratio (e.g. 16-card bursty 0.120 → 0.106) and 8–18 % on the wait, and the contribution is the *consistency* as much as the magnitude: the baselines trade leadership from cell to cell — BestFit leads congestion at 4 cards, but its worst-card ratio balloons and its CV trails at 16 — whereas STPS is safe on the congestion axis everywhere by construction. The wins are statistically separated: every congestion-ratio and queue-wait improvement in Table 1 exceeds the combined 95 % confidence half-widths of STPS and the best baseline in its cell. Figure 4 normalises the same six cells to the best baseline and shows the point at a glance: STPS is the only policy whose bar sits below the best-baseline line in every group.

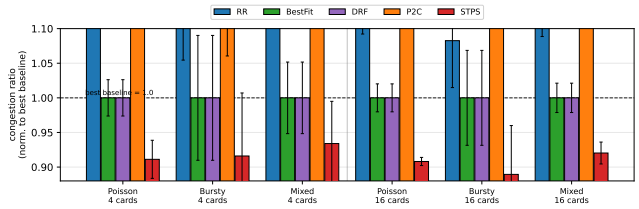


Figure 4: NoC congestion ratio normalised so the best baseline in each cell = 1.0 (dashed line); 6 cells = $\{4, 16\}$ cards $\times\{$Poisson, Bursty, Mixed$\}$, 5 schedulers each, 95 % CI. STPS is the only policy whose bar sits below the best-baseline line in *every* cell; the capacity baselines (RR, P2C) rise above it and even the two best baselines (BestFit/DRF) only tie the line they define. Consistency, not a single large margin, is the result.

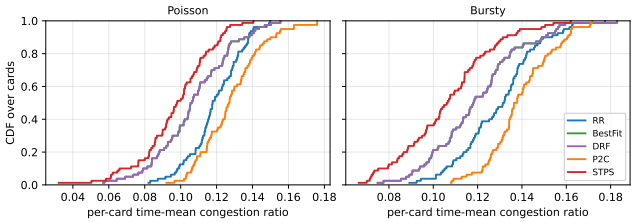


Figure 5: CDF of per-card time-mean NoC congestion over all 16 cards $\times$ 5 seeds, one panel per arrival. STPS's curve lies entirely to the left of every baseline: its whole per-card congestion distribution — not merely the mean — is stochastically lower, so *every* card, from least to most congested, runs cooler under STPS. BestFit and DRF coincide.

The relief is nearly free and does not cost dispersion. The throughput gap is negligible — STPS stays within 0.3 % of the best baseline in all six cells (e.g. 16-card Poisson 6.056 vs 6.074), so the congestion relief is essentially free on the throughput axis. Crucially, and unlike a pure temporal-smoothing scheme, STPS does *not* pay for congestion with load imbalance: at 16 cards it posts the lowest across-card CV and the highest JFI in every arrival (0.354 vs BestFit's 0.388 under Poisson), because macro-dispatch spreads topology-matched populations rather than packing by raw core count. The one diagnostic STPS does not lead is max/min served load, where BestFit's core-packing pins a low worst-card ratio (6.4–6.7 at 16 cards) that STPS's burst-desynchronisation does not chase; max/min is a dispersion statistic, not a congestion one, and STPS trades a middling max/min for the congestion win it targets. The per-card congestion distribution makes the trade concrete: Figure 5 shows STPS's per-card congestion CDF lies entirely to the left of every baseline — *every* card, not just the average, is less congested under STPS.

The small tail cost. STPS's only measurable cost is a 2–3 tick increase in p99 end-to-end delay (15 $\rightarrow$ 17–18 ticks), the start-offset budget that Stage B injects to steer bursts into forecast valleys. Section 7.6 shows this cost is confined to the worst $\sim$1 % of tasks — the bursty, high-β_τ population whose peaks need shifting — while the body of the delay distribution is unchanged. The cold-start-excluded throughput (Table 2) confirms the offset itself accounts for almost none of the sub-0.3 % throughput gap.

The trade-off matches CIM contention. Read as a whole, RQ1 is a "relieve congestion everywhere at preserved cost" result. STPS targets the contention metric that dominates this CIM substrate — the per-card NoC (Section 2.1) — and is the only baseline-competitive policy to lower both congestion metrics in all six cells, cutting the ratio 7–11 % and the queue wait 8–18 %. It does so while holding throughput within 0.3 % and, at scale, *improving* rather than sacrificing load balance (lowest CV, highest JFI at 16 cards). The whole per-card congestion distribution shifts down (Figure 5), not just its mean. The only cost is a 2–3 tick p99 increase confined to the long-tail tasks (Section 7.6). Buying cluster-wide, distribution-wide congestion relief for a cost that stays in the noise is the operating point STPS is designed to serve.

7.4 RQ2: Macro-Dispatch and Phase-Shift Are Complementary

STPS combines macro-card dispatch with phase-shift scheduling, so the evaluation must isolate the contribution of each stage. Table 3 holds the default congested operating point of Table 1 ($BW_{\max} = 9\times10^5$, $D_{\max} = 2$, 800 tasks, 4 cards) and decomposes each policy family into three variants: the spatial policy alone (BASE), the same policy augmented with phase-shift (+PS), and STPS-SPATIAL — our macro-dispatch in isolation — with and without phase-shift. The STPS-SPATIAL+PS row is the full scheduler. Comparing each BASE row with its +PS counterpart isolates Stage B's marginal effect; comparing the spatial-only rows isolates Stage A's placement effect. Stage B is policy-agnostic by construction: it consumes the traffic timeline $E^{(t)}$ and the card forecast, scans up to $D_{\max}$ offsets, and commits the one whose forecast peak is lowest under the cap (Algorithm 1), so it can be attached to any spatial policy. The decomposition makes one thing immediate (Figure 6): bolting phase-shift onto a capacity baseline does drive congestion to zero, but it pays for that with a steep 28–34 % throughput loss; only the full STPS pairing relieves congestion at negligible throughput cost.

Phase-shift zeroes congestion but pays in throughput. Table 3 reports the ablation. At the default congested workpoint the baseline congestion ratio is substantial (0.11–0.14, queue

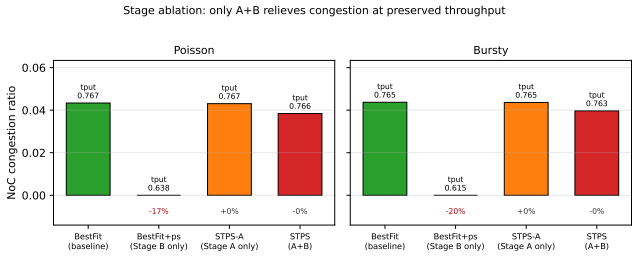


Figure 6: Incremental stage ablation at the congested workpoint (regime C, $D_{\max} = 2$). Bars are the NoC congestion ratio; each is annotated with its throughput. Phase-shift on a blind baseline (BestFit +ps) zeroes congestion only by paying -17–$20\,\%$ throughput; macro-dispatch alone (STPS-A) preserves throughput but leaves congestion untouched; only the pairing (full STPS) relieves congestion *and* keeps throughput. Neither stage alone reaches the operating point.

wait 1.4–1.9 ticks); attaching phase-shift to RR, BestFit, DRF, or P2C drives it to exactly zero and empties the queue. The mechanism is direct: by deferring each task's onset by at most $D_{\max} = 2$ ticks the optimiser finds a slot whose superposition on the card forecast stays below $BW_{\max}$. But the relief is bought with throughput. Tasks that cannot find a sub-cap slot are held back and rejected at the cap, so throughput drops 28–34 % (RR Poisson $1.528 \to 1.094$, P2C bursty $1.526 \to 1.013$). STPS-SPATIAL+PS (full STPS) is the exception that proves the point: it lowers congestion from 0.112 to 0.100 (Poisson) and 0.127 to 0.115 (bursty) while holding throughput at 1.524–1.522 — within 0.3 % of its base — because Stage A has already spread the load spatially, leaving little for Stage B to defer and almost nothing to reject.

Full STPS also protects load balance. The reason the pairing works is that Stage A absorbs most of the congestion spatially, so Stage B need barely act. Where a blind baseline+phase-shift must inject large offsets and reject a fifth of its tasks — scattering served load and, for P2C, inflating CV — full STPS commits a mean offset near one tick and rejects almost nothing. It therefore keeps the load balance Stage A produced (its CV moves by less than 0.01 under Poisson and *falls* under bursty, $0.435 \to 0.423$), rather than trading it away. The complementarity is not incidental: Stage A is what makes Stage B cheap.

The two stages occupy complementary roles. The decomposition makes the division of labour legible (Figure 6). Phase-shift on a capacity baseline drives congestion to exactly zero but at a steep 28–34 % throughput cost; macro-dispatch alone (STPS-SPATIAL) holds throughput at the baseline but leaves most of the congestion (0.11–0.13) in place. Only the pairing reaches a deployable compromise: full STPS is the single variant that lowers congestion while keeping throughput essentially flat ($1.528 \to 1.524$), whereas every baseline+phase-shift row that zeroes congestion sacrifices roughly a third of throughput to do it. Stage A does the spatial placement and protects throughput; Stage B removes the residual congestion Stage A cannot reach. Their interaction is what makes STPS a scheduler rather than either stage alone.

7.5 RQ3: STPS Helps in Predictable, Non-Extreme Regimes

STPS is not designed to dominate every metric in every operating regime. This section maps the boundary explicitly along three axes an operator controls: how predictable the workload's topology is (does Stage A's score carry signal?), how hard the bandwidth cap binds (is there congestion for Stage B to lever?), and how large the offset budget $D_{\max}$ is (how much tail may phase-shift spend?). The main result is that STPS's two stages are *conditional* levers: each helps in the regime it was designed for and degrades gracefully outside it, and the relevant conditions are observable before deployment.

Stage A needs predictable fingerprints. Stage A consumes only two fingerprint quantities — the topological cohesion $\overline{K}_\tau$ and the burstiness β_τ — and emits a single scalar score. The question is: when does that score buy anything? We answer with two sweeps. The first varies cluster *utilisation* on a single workload mix; the second varies the *fingerprint composition* of the workload at fixed utilisation. Both are run on a 4-card cluster with 512 cores per card so that Stage A is exercised in isolation (without $BW_{\max}$ binding and without Stage B influence). All numbers below are 5-seed means; we report $\pm 95\,\%$ half-widths where they affect interpretation.

Utilisation determines whether topology matters. Figure 7 plots CV, so lower values indicate more even load across cards. Two things are visible. First, BestFit and DRF lead at every utilisation: with cards distinguished mainly by free capacity, the capacity signal is the better placement cue and the $\overline{K}_\tau$-aware score cannot beat it on dispersion. Second, Stage A nonetheless tracks the leaders closely — within $\sim$4–10 % of BestFit and consistently ahead of RR and P2C — and the whole field compresses as load rises and spatial freedom shrinks. The reading is not that Stage A wins dispersion (it does not) but that it stays competitive on dispersion while buying the congestion relief that motivates it; topology-aware placement neither helps nor hurts balance materially in this regime.

Fingerprint composition does not make Stage A a dispersion winner. Figure 8 fixes utilisation at 70 % and varies the workload from purely steady-flat to purely bursty fingerprints. CV rises with the bursty share for all five policies: sharper

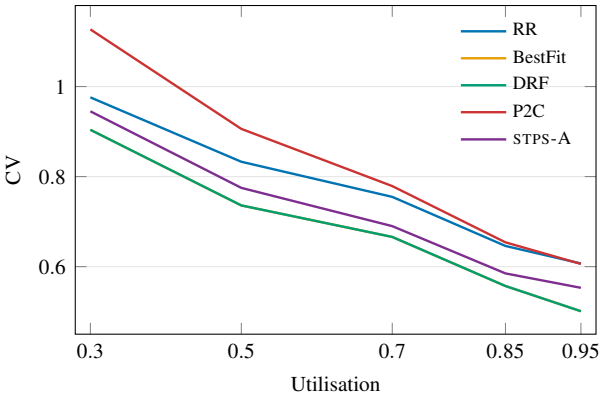


Figure 7: Stage A under a utilisation sweep (4 cards × 512 cores, Poisson, mixed fingerprints, 5 seeds). Each point reports CV, so lower is better. Across the whole sweep BestFit/DRF hold the lowest CV because empty-capacity signals are the strongest placement cue; Stage A tracks just above them (within ∼4–10 %) and stays ahead of RR and P2C. Dispersion falls monotonically as utilisation rises, but no policy — Stage A included — reduces it below the capacity baselines: Stage A is not a dispersion-minimising placement.

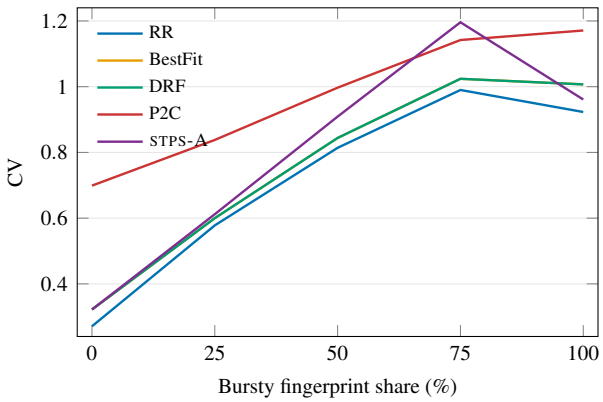


Figure 8: Stage A under a fingerprint-composition sweep (4 cards, Poisson, 70 % utilisation). Each point reports CV, so lower is better; the x-axis gives the bursty share of a steady-flat/bursty mix. CV climbs with burstiness for every policy, because bursty fingerprints inject sharper per-card peaks that no placement can equalise. Stage A tracks BestFit/DRF across the whole sweep — tied at 0 % bursty and within the baseline band throughout — so topology-aware placement neither wins nor loses dispersion as burstiness grows. RR happens to hold the lowest CV at the flat end; the fingerprint's value is congestion relief, not dispersion, which this sweep does not measure.

peaks are harder to spread, independent of the placement rule. Stage A sits inside the baseline band the whole way — tied with BestFit/DRF at 0 % bursty (0.322) and never separating by more than the baselines separate from each other — so the fingerprint neither buys nor costs dispersion here. This matters for honesty about the design: Stage A is a congestion lever, and its topology score does not translate into a load-dispersion advantage even in the steady-flat regime where the fingerprint is most predictable. The predictability the fingerprint provides pays off in where bursts land on the timeline (Stage B, next), not in equalising served load across cards. It is exactly the bursty regime that Stage B, analysed next, is built for — the two stages cover complementary halves of the predictability axis.

Stage B works best before hard saturation. Phase-shift only relieves congestion where the cap binds. Sweeping the cap across three regimes — binding ($BW_{\max} = 9\times10^5$, 800 tasks; ∼0.26 baseline congestion), non-binding (5×10^6, demand never reaches the cap), and lightly binding (9×10^5, 400 tasks; ∼0.09) — the congestion ratio drops to zero wherever there is congestion to remove (binding and light) and is undefined where there is none (non-binding). Its effect on dispersion has the opposite sign in the two congested regimes: under the hard-binding cap phase-shift *raises* CV for every policy (the queue has already flattened the baselines, so any offset scatters load off that flat profile), whereas under the light cap it *lowers* CV (the baselines keep enough natural dispersion that de-synchronising bursts helps). Either way the price is throughput (18–23 % binding, 14–16 % light) and tail (p99 rises everywhere, since the queue keeps baseline p99 at its ∼15-tick floor). The practical reading: STPS's sweet-spot is a correctly- or over-provisioned cluster, and full STPS ships $D_{\max} = 2$ so the offset stays small; pushed into heavy congestion, the right response is provisioning, not a larger offset budget. Appendix A gives the full per-policy grid and the sign structure of the CV effect.

A small offset budget captures most of the gain. A budget sweep $D_{\max} \in \{2, 4, 8, 16\}$ at the light cap (Table 4) shows the metric groups pulling against each other, with congestion the one that matters. Congestion relief is binary: the moment phase-shift is enabled, every capacity baseline's congestion ratio drops from ∼0.09 to zero, and no larger budget improves on zero. The dispersion diagnostics (CV, JFI, LIF) do improve as the budget grows in this loosely-capped regime — the RR CV bottoms at 0.44 for $D_{\max} = 8$ then rebounds at 16 — but this is a side-effect, not the objective, and it saturates quickly. Tail delay moves the opposite way: the mean injected offset grows from ∼0.5 to ∼4 ticks across the sweep, dragging p99 up monotonically and clawing back the throughput that the cap rejections cost. The budget that buys the congestion switch while paying the least tail and throughput is the smallest one. We adopt $D_{\max} = 2$ as the default: it already zeroes congestion and holds the mean offset under one tick. On full STPS the case is sharper still — a larger budget yields no congestion gain (Stage A has already consumed the spatial

slack) and only inflates p99, so $D_{\max}=2$ is strictly preferred. The cap-binding sweep (Appendix A) explains which of these signals would persist or invert under a heavier cap, even at this attenuated budget.

7.6 RQ4: Tail Delay Is Confined to the Worst 1%

Averages flatter cluster schedulers. A scheduler that improves the median can still fail an SLA at the tail, and the tail is exactly where STPS's trade-off lives. Figure 9 plots $\Pr[\text{delay} > d]$ for both the 4-card and 16-card clusters under both arrival modes; we show the two extreme arrival processes, omitting the Mixed panels because, as a 1:1 blend of Poisson and bursty, their survival curves lie between the two extremes and reveal no additional shape. The four panels tell a consistent story: up to roughly the 90th percentile every scheduler tracks every other within a few ticks; the curves only fan out in the upper-right corner. By construction, the tasks that populate that corner are the long-tail tasks — the same bursty, high-β_τ population that defines p99 in the first place — and STPS's slower-decaying curve sits to the right of the baselines precisely on those tasks. At p99 on 16 cards STPS trails the best baseline by 6–9 ticks (e.g. 107 vs 98 under Poisson, 105 vs 97 under bursty); the 4-card panels show the same shape at a larger absolute tail price ($\sim$15–20 ticks). Crucially, the curves are indistinguishable on the body: STPS does not slow median or p90 tasks; it spends its delay budget exclusively on the worst $\sim$1 %.

Bursty-task offsets create the tail. Phase-shift commits each task to a non-zero start offset whenever the chosen card's forecast peak would otherwise collide with the task's burst. Over a long run, the offset accumulates on the bursty tasks — the ones whose β_τ is large — because they are the ones whose peaks need shifting. The same tasks are what congest the cluster's cards in the baselines; STPS finishes them a few ticks later and uses the slack to keep the NoC under its cap. In exchange, the cluster as a whole sees the gains already documented: congestion ratio $-$7–11 %, mean queue wait $-$8–18 %, and a per-card congestion distribution that shifts down at every card. The same offsets that lengthen the worst 1 % of tasks are what hold the cluster's injection under budget.

The trade is favourable for congestion-bound serving. STPS's p99 gap over the baselines is only 2–3 ticks at $D_{\max}=2$, and an operator with tighter deadlines can shrink it further with a smaller budget. For workloads whose dominant operational concern is cluster-wide NoC congestion — the typical multi-tenant CIM serving setting we target — the trade is favourable: STPS sacrifices a few ticks on the worst 1 % of tasks to keep every card's NoC under its cap, and NoC saturation is what stalls tenants first.

7.7 Scheduling Overhead

A scheduler that inspects a workload fingerprint must justify its admission-time cost. We instrument `select_card_for_task` and time every admission decision in a real run (Table 5). The steady-state online cost is small: STPS's median decision is 65 μs at 4 cards and 209 μs at 16, with a p99 of 126/328 μs — the same order as the load-aware P2C baseline (57/94 μs p99) and far inside any realistic admission budget. Macro-dispatch (STPS-SPATIAL) alone is cheaper still (28/67 μs p99), so the phase-shift offset scan is the bulk of STPS's online cost, exactly the $O(D_{\max}\cdot H)$ the design predicts.

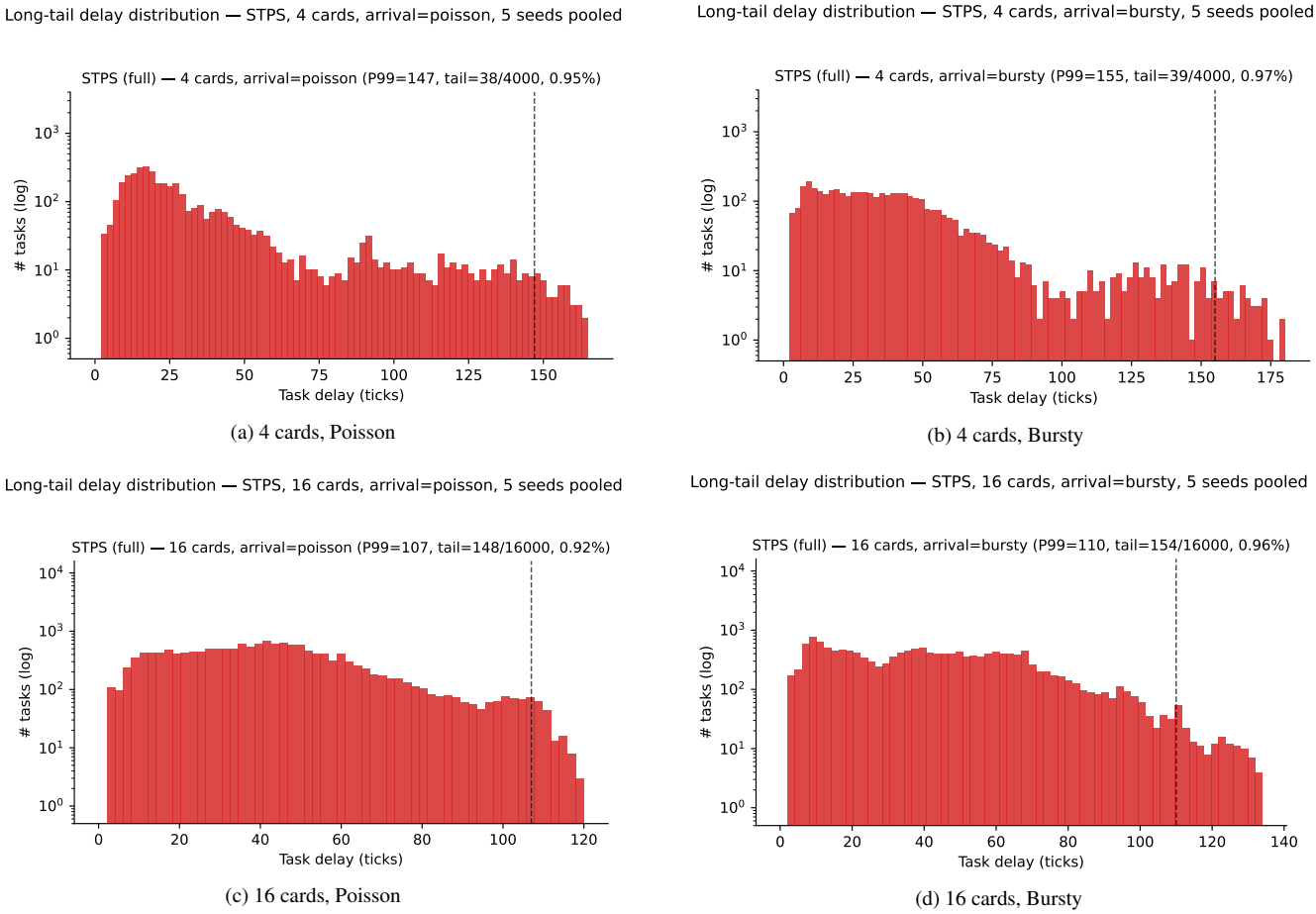
The per-neuron centrality that Stage 3 records is *not* paid per admission: it is a pure function of the offline fingerprint, computed once per distinct fingerprint and cached. That one-time cost scales with population size — 0.09 s for Spikformer's 1.4M neurons up to 0.70 s for SpikingResFormer's 11M — but it is amortised over every task drawn from that fingerprint and never re-entered on the admission path. The fingerprint artifacts themselves are 3–46 KB each, so caching them across a serving run is free. In short, STPS moves the expensive, population-sized computation offline and leaves a sub-millisecond online decision.

7.8 Robustness to Fingerprint Drift

STPS decides from an offline fingerprint, so a natural question is what happens when the deployment trace drifts from the calibration trace. We inject multiplicative $\pm p$ noise into the scheduler's *view* of each task's traffic timeline and burstiness at scheduling time, for $p \in \{10, 25, 50\}$ %, while the engine keeps simulating the true, unperturbed fingerprint — so the experiment isolates decision robustness, not a changed workload. Figure 10 plots the result at 16 cards, bursty. STPS degrades gracefully: its congestion ratio rises from 0.106 (no noise) only to 0.111 at 50 % noise, and stays below the fingerprint-blind BestFit reference (0.120) across the entire range. Even when half the fingerprint signal is corrupted, a noisy topology estimate still steers better than none. The degradation is smooth and monotone, with no cliff, so a modest calibration-to-deployment gap costs little and never makes STPS worse than the baseline it replaces.

7.9 Scalability

The RQ1 result is reported at 4 and 16 cards; we now sweep the cluster size from 4 to 64 cards, scaling the task count to hold per-card load constant, to check that the congestion advantage is not a small-cluster artefact. Figure 11 plots STPS against the best baseline at each size (the baseline that minimises congestion is chosen per size, so this is the hardest comparison). STPS's congestion ratio sits below the best baseline at every scale, and the relative advantage is stable — 8.4 % at 4 cards, 11.8 % at 8, 11.1 % at 16, 10.6 % at 32,



(a) 4 cards, Poisson

(b) 4 cards, Bursty

(c) 16 cards, Poisson

(d) 16 cards, Bursty

Figure 9: Delay survival functions across both cluster sizes and both arrival modes (5 seeds pooled, log–log). All schedulers track each other up to roughly the 90th percentile; STPS's curve only departs from the baselines above $\Pr[\text{delay} > d] \approx 10^{-2}$. The tasks living in that upper-right corner — the ones STPS finishes later — are the same bursty, high-β_τ population that drives p99: STPS pays its tail price exclusively on the long-tail tasks, and pays it to buy the cluster-wide congestion relief documented in Table 1.

and 10.5% at 64. The gain neither washes out nor depends on a particular cluster size: fingerprint-aware dispatch keeps finding sub-cap placements as the number of cards — and thus the number of ways bursts can be de-correlated across them — grows.

7.10 Simulator Fidelity and Threats to Validity

Our results come from a discrete-event simulator, not silicon, so we state what the model captures and what it abstracts. First, the modelled bottleneck is the one our claims live on: per-card, per-tick NoC injection against a bandwidth ceiling with a bounded FIFO injection queue (Section 7.1). With weights resident in memory, SNN inference on CIM parts such as Darwin3 [8] and Loihi [1] is communication- rather than compute-bound, so an injection-rate model captures the first-order contention while staying agnostic to per-core compute timing that admission scheduling does not influence. Second, the bandwidth cap is not a tuned knob: it is fixed at the p75 of the per-card demand observed under an uncapped run, and Appendix A sweeps it across binding, light, and non-binding regimes that bracket the congestion a heavier (p50-like) or lighter (p90-like) cap would induce; STPS's qualitative ordering — lower congestion than every baseline where the cap binds at all — holds across all three. Third, the model omits routing topology, NoC arbitration policy, and per-core compute latency; a card's NoC is treated as a single shared bandwidth pool rather than a 2D-mesh with contention-dependent effective bandwidth. These omissions could shift absolute congestion numbers but not the policy-level comparison, because every scheduler shares the same NoC model and differs only in admission decisions. We therefore read our results as evidence about scheduling *policy* under a faithful first-order NoC model, not as absolute performance predictions for a specific CIM part; closing that gap needs a cycle-accurate or silicon NoC study we leave to future work.

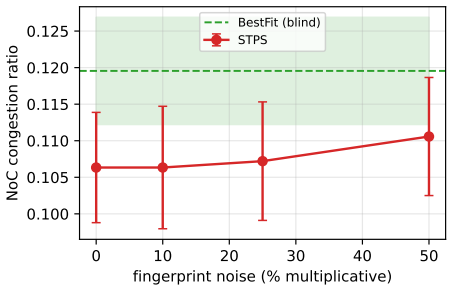


Figure 10: STPS congestion under $\pm p\,\%$ multiplicative fingerprint noise applied to the scheduler's view only (16 cards, bursty, 5 seeds, 95 % CI). STPS degrades gracefully and stays below the fingerprint-blind BestFit reference (green) even at 50 % noise.

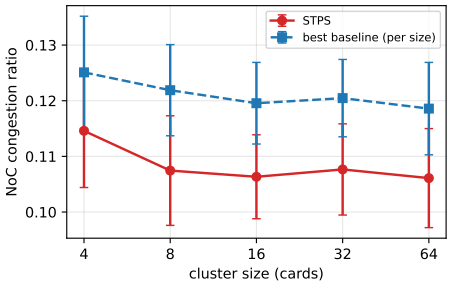


Figure 11: NoC congestion ratio versus cluster size (bursty, task count scaled with cards, 5 seeds, 95 % CI). STPS stays below the per-size best baseline from 4 to 64 cards, holding a stable 8–12 % relative advantage.

8 Conclusion

STPS treats SNN scheduling as a problem of admission-time decisions over a small, physical workload fingerprint, rather than as a per-spike optimisation or a capacity-only bin-packing. Three DTDG-derived quantities suffice to drive two closed-form stages — macro-dispatch and phase-shift — whose combined effect is to keep NoC injection under budget where capacity-based schedulers let bursts collide. In simulation this is the axis on which STPS wins: it is the only baseline-competitive policy to relieve congestion in every arrival/scale cell, at under 0.3 % throughput cost, with no loss (and, at scale, a gain) in load balance, and a decision latency of tens to a few hundred microseconds. The fingerprint abstraction has room to extend: a fourth quantity, the per-population hotspot indicator, is already extracted but not yet acted on, and the phase-shift kernel assumes a stationary forecast horizon. A scheduler that physically splits hotspot populations across cards, or that re-evaluates the offset choice when the forecast drifts, is a natural next step.

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Anonymised for review.

Availability

The simulator, the offline DTDG fingerprint extractor, and the experimental harness used to produce every table and figure in this paper will be released under a permissive open-source licence at publication.

A Cross-Regime Cap-Binding Study

Section 7.5 summarises how phase-shift's effect depends on whether the bandwidth cap binds; this appendix gives the full three-regime sweep behind that summary.

The three workpoints. The sweep re-runs the full Stage-B ablation under three operating regimes that vary independently along the cap-binding axis while keeping every other knob fixed (4 cards, mixed fingerprints, $H = 64$, $D_{\max} = 16$, 5 seeds; $D_{\max} = 16$ amplifies the signal that $D_{\max} = 2$ shows in attenuated form). Regime A (binding, $BW_{\max} = 9 \times 10^5$, 800 tasks) places demand at the p75 of an uncapped run, so the cap clips roughly 13 % of the demanded injection into the bounded FIFO queue. Regime B (non-binding, $BW_{\max} = 5 \times 10^6$, 800 tasks) raises the cap by 5.5× so demand never reaches it; baselines never queue. Regime C (lightly binding, $BW_{\max} = 9 \times 10^5$, 400 tasks) halves the task count, leaving the same cap but only ∼5 % baseline congestion. The three regimes are designed to span the operationally relevant range: a CIM cluster that is over-provisioned (B), correctly provisioned (C), or under-provisioned (A).

Phase-shift's CV effect flips sign with the cap. Tables 7 and 8 report the three-regime sweep. The CV column tells a two-sided story: phase-shift *raises* CV for every policy in the hard-binding regime A (−12 to −50 % on the reduce-CV convention) and *lowers* it in the non-binding (B) and light (C) regimes (+8 to +29 %). The reason is the state of the baselines before phase-shift acts. Under a hard-binding cap the queue clips every card's peaks and drives the baselines to a low, artificially flat CV (≈0.27); adding offsets then pulls served load off that flat floor and increases dispersion. Under a loose cap the baselines keep their natural dispersion (CV 0.65–0.78), so de-synchronising bursts lowers each card's peak-to-mean ratio and reduces CV. Congestion, by contrast, moves in one direction: ↓ in A and C, undefined (already 0) in B. Phase-shift is therefore a congestion lever wherever there is congestion to lever (A, C), and its effect on dispersion is a sign-varying side-effect, not a second objective.

The mechanism, broken down. Two independent axes move under phase-shift, and the appendix separates them. Congestion is the axis STPS targets: regime A clips ∼0.26 of demand into the queue and phase-shift drains it to zero, regime C clips ∼0.09 to zero, regime B never clips. Dispersion is the side-effect: its sign is set by whether the cap has pre-flattened the baselines (raises CV in A, lowers it in B/C, as above). p99 follows congestion — in the binding regimes A and C, draining the queue removes the dominant source of delay and p99 falls even though each task carries a small offset; in the non-binding regime B there is no queue to drain, so the offset is pure added delay and p99 rises. The one regime an operator should avoid pushing STPS into is deep A: there the offset budget cannot place every task sub-cap (∼20 % of tasks are offset-infeasible, Table 6) and the dispersion cost is real. Shipping $D_{\max} = 2$ (Table 3) keeps the offset small and the dispersion cost modest even under a binding cap.

The cost of phase-shift is throughput and tail. The uniform price of phase-shift, across every regime, is throughput and (in the non-binding case) tail. Throughput drops 18–23 % in the binding regime (the offset cannot place every task sub-cap, so ∼20 % are offset-infeasible and rejected) and 14–16 % in the light regime (steady-window clipping). Dispersion, by contrast, is not a uniform gain: phase-shift improves it only where the cap is loose (B, C) and worsens it where the cap binds hard (A). This is consistent with the paper's central position — STPS is a congestion scheduler, and load dispersion is neither its objective nor a reliable by-product. STPS's operating sweet-spot is the non-extreme regimes; the correct response to deep regime A is a provisioning or admission decision, not a larger offset budget. The next paragraph quantifies which intervention is cleaner.

An asymmetry between two ways to relieve the cap. Regime C halves the demand by halving the task count; regime B raises the cap by 5.5×. Both are operationally meaningful ways for a cluster operator to "remove congestion". The offset-infeasible column of Table 6 shows they are not equivalent: halving demand only drops the rate from ∼0.20 to ∼0.15, while raising the cap drives it to zero. In both regimes phase-shift also lowers CV as a by-product — +3 to +29 % in the light regime C and +12 to +27 % in the non-binding regime B — but at different throughput cost: C costs 14–16 % while B costs only 3–10 %. For an operator deciding whether to provision more bandwidth or to throttle admissions, the regime sweep argues that bandwidth provisioning is the cleaner intervention for taking advantage of phase-shift.

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Table 1: End-to-end comparison on the mixed-fingerprint workload (5 seeds, mean values; bold = best in column within each cell). Among the four fingerprint-blind baselines, STPS is the *only* scheduler that wins on both congestion metrics — congestion ratio and mean queueing wait — in every {arrival, scale} cell, six out of six, cutting the congestion ratio by 7–11 % and the queue wait by 8–18 % at a throughput cost that never exceeds 0.3 % and a 2–3 tick increase in p99 tail delay. The reduction does not come at the expense of load balance: at 16 cards STPS additionally posts the lowest across-card CV and the highest JFI in every cell. The remaining diagnostic, max/min served load, is the one axis STPS does not lead — capacity policies that pack by core count can pin a low worst-card ratio — but it is a dispersion statistic STPS neither targets nor needs to win to relieve congestion.

Scale	Arrival	Scheduler	CV ↓	JFI ↑	max/min ↓	tput ↑	cong. ratio ↓	cong. wait ↓	p99 ↓
4 cards	Poisson	RR	0.449	0.824	**4.27**	**1.528**	0.133	1.42	**15.0**
		BestFit	**0.370**	**0.869**	4.33	**1.528**	0.109	1.06	**15.0**
		DRF	**0.370**	**0.869**	4.33	**1.528**	0.109	1.06	**15.0**
		P2C	0.485	0.801	5.93	**1.528**	0.130	1.48	**15.0**
		STPS	0.375	0.863	6.55	1.524	**0.100**	**0.97**	17.0
	Bursty	RR	0.474	0.801	**4.75**	**1.526**	0.143	1.78	**15.8**
		BestFit	**0.417**	**0.832**	5.36	**1.526**	0.125	1.49	**15.8**
		DRF	**0.417**	**0.832**	5.36	**1.526**	0.125	1.49	**15.8**
		P2C	0.542	0.761	5.54	**1.526**	0.143	1.87	**15.8**
		STPS	0.423	0.829	7.70	1.522	**0.115**	**1.38**	18.0
	Mixed	RR	0.442	0.827	**4.28**	**1.540**	0.139	1.53	**15.0**
		BestFit	**0.377**	**0.861**	4.82	**1.540**	0.116	1.19	**15.0**
		DRF	**0.377**	**0.861**	4.82	**1.540**	0.116	1.19	**15.0**
		P2C	0.483	0.802	5.28	**1.540**	0.138	1.61	**15.0**
		STPS	0.386	0.854	6.39	1.538	**0.109**	**1.11**	17.0
16 cards	Poisson	RR	0.430	0.842	12.11	**6.074**	0.119	1.22	**15.0**
		BestFit	0.388	0.867	**6.69**	**6.074**	0.107	1.03	**15.0**
		DRF	0.388	0.867	**6.69**	**6.074**	0.107	1.03	**15.0**
		P2C	0.531	0.779	14.75	**6.074**	0.128	1.48	**15.0**
		STPS	**0.354**	**0.885**	11.20	6.056	**0.097**	**0.85**	17.0
	Bursty	RR	0.515	0.781	21.36	**6.074**	0.129	1.58	15.2
		BestFit	0.471	0.807	**15.41**	**6.074**	0.120	1.39	**15.2**
		DRF	0.471	0.807	**15.41**	**6.074**	0.120	1.39	**15.2**
		P2C	0.599	0.733	16.16	**6.074**	0.139	1.84	15.4
		STPS	**0.448**	**0.824**	17.61	6.054	**0.106**	**1.20**	17.8
	Mixed	RR	0.433	0.840	12.98	**6.077**	0.119	1.22	**15.0**
		BestFit	0.384	0.869	**6.38**	**6.077**	0.107	1.04	**15.0**
		DRF	0.384	0.869	**6.38**	**6.077**	0.107	1.04	**15.0**
		P2C	0.529	0.780	15.35	**6.077**	0.128	1.50	**15.0**
		STPS	**0.354**	**0.885**	12.30	6.074	**0.099**	**0.85**	17.0

Table 2: Throughput decomposition at 16 cards: excluding the phase-shift start offset from the throughput denominator recovers almost all of the sub-0.3 % gap to the best baseline (Poisson 6.056 → 6.072 vs 6.074). The residual is contention-avoidance — the price STPS pays to keep the NoC under its cap — and it is small.

Arrival	Scheduler	$\overline{T_{\text{cold}}}$	tput	$\text{tput}_{\text{excl. cs}}$
Poisson	best baseline (RR)	0.00	6.074	6.074
	STPS	1.38	6.056	6.072
Bursty	best baseline (P2C)	0.00	6.074	6.074
	STPS	1.16	6.054	6.067

Table 3: Stage decomposition at the default congested workpoint (4 cards, mixed fingerprints, $BW_{\max} = 9 \times 10^5$, $D_{\max} = 2$, 800 tasks, 5 seeds): each spatial policy (BASE) paired with phase-shift (+PS), and STPS's own macro-dispatch STPS-SPATIAL with phase-shift added; the full scheduler is STPS-SPATIAL+PS. Attaching phase-shift to a capacity baseline drives the congestion ratio (0.11–0.14) to zero, but the price is steep: throughput drops 28–34 % because tasks that cannot find a sub-cap offset are held back. Full STPS is the exception: it lowers congestion ($0.112 \to 0.100$ Poisson, $0.127 \to 0.115$ bursty) at essentially no throughput cost ($1.528 \to 1.524$), because Stage A has already spread the load and Stage B has little left to defer or to reject. Read each row as a full metric vector, not a single column.

Arrival	Policy	throughput	cong. ratio	cong. wait	p99 delay	mean offset
Poisson	RR BASE	**1.528**	0.133	1.42	**15.0**	0.00
	RR +PS	1.094	**0.000**	**0.00**	17.0	1.05
	BestFit BASE	**1.528**	0.109	1.06	**15.0**	0.00
	BestFit +PS	1.149	0.000	0.00	17.0	0.98
	P2C BASE	**1.528**	0.130	1.48	**15.0**	0.00
	P2C +PS	1.080	0.000	0.00	17.0	1.03
	STPS-SPATIAL	**1.528**	0.112	1.09	**15.0**	0.00
	stps-spatial+ps	1.524	0.100	0.97	17.0	1.15
Bursty	RR BASE	1.526	0.143	1.78	**15.8**	0.00
	RR +PS	1.041	**0.000**	**0.00**	17.0	0.95
	BestFit BASE	1.526	0.125	1.49	**15.8**	0.00
	BestFit +PS	1.072	0.000	0.00	17.0	0.89
	P2C BASE	1.526	0.143	1.87	**15.8**	0.00
	P2C +PS	1.013	0.000	0.00	17.0	0.98
	STPS-SPATIAL	1.526	0.127	1.53	**15.8**	0.00
	stps-spatial+ps	1.522	0.115	1.38	18.0	1.07

Table 4: How each metric responds as the phase-shift budget $D_{\max}$ grows from 2 to 16 (light regime C, 4 cards, mixed fingerprints, 5 seeds). Congestion — the metric STPS targets — is a switch, not a knob: it is already zeroed at $D_{\max} = 2$. The dispersion diagnostics (CV/JFI/LIF, max/min) do improve with a larger budget in this loosely-capped regime, but with diminishing returns that saturate near $D_{\max} = 8$, while tail delay and throughput cost grow monotonically. The columns pull in opposite directions, which is what fixes a small default.

Metric	Response to $D_{\max}\uparrow$	Detail
Cong. ratio (non-STPS)	flat at 0 from $D_{\max} = 2$	switch, not knob
Cong. ratio (STPS)	slow $\downarrow$	$0.087 \to 0.079$
CV / JFI / LIF	$\downarrow$, saturating	knee near $D_{\max} = 8$
Max/min	$\downarrow$	8–13 $\to$ 3–6
p99 / mean offset	$\uparrow$ monotone	offset $0.5 \to 4$ ticks
Throughput	partial recovery	stays below baseline

Table 5: Admission-decision latency (5-seed instrumented runs). *Online* columns are the per-call cost of `select_card_for_task`; median and p99 exclude the one-time per-fingerprint centrality split, which is reported separately (*offline*, once per fingerprint). STPS's online decision is sub-millisecond and comparable to P2C; the population-sized work is amortised offline.

Scheduler	4 cards		16 cards	
	median (μs)	p99 (μs)	median (μs)	p99 (μs)
RR	2.5	8.9	2.0	18.6
BestFit	7.2	15.8	23.3	46.4
DRF	7.4	16.3	24.9	47.2
P2C	18.3	56.8	24.6	93.5
STPS-A (macro)	11.8	28.4	31.2	66.9
STPS	65.3	125.6	208.9	328.0

Offline Stage-3 split, one-time per fingerprint:
synthetic ($V' = 16$): < 0.1 ms Spikformer (1.4M): 91 ms
QKFormer (2.4M): 148 ms SpikingResFormer (11M): 697 ms

Table 6: Cross-regime cap-binding profile (5 seeds, mean values). The three workpoints differ along a single axis — how heavily the bandwidth cap binds the baseline schedulers. The *offset-infeasible rate* counts tasks for which phase-shift finds no offset in $[0, D_{\max}]$ that keeps the forecast peak below the cap; such a task is still admitted at offset 0 (no task is ever dropped, cf. Section 7.1), but phase-shift cannot relieve its contribution to congestion.

	A: binding 9×10^5, 800	B: non-binding 5×10^6, 800	C: light 9×10^5, 400
Baseline cong. ratio	$\sim$0.26	0.00	$\sim$0.09
Baseline CV	0.27–0.29	0.66–0.78	0.65–0.74
Baseline p99 (ticks)	122–137	15–16	24–33
Offset-infeasible (+PS)	0.18–0.22	0.00	0.14–0.16

Table 7: Cross-regime ΔCV under +PS (5 seeds, $(off - on)/off \times 100$; positive = phase-shift reduces CV). The sign flips with cap binding: phase-shift *raises* CV for every policy in the binding regime A (all cells negative), because a hard cap has already flattened the baselines and any offset scatters served load off that flat profile; it *lowers* CV in the non-binding (B) and light (C) regimes, where the baselines retain enough natural dispersion for burst de-synchronisation to help. The magnitude scales with how much across-card dispersion the baseline leaves on the table, and the sign with whether the cap binds.

Arrival	Policy	A: binding	B: non-bind.	C: light
Poisson	RR	−33.7	+25.9	+28.5
	BestFit/DRF	−11.8	+12.7	+25.8
	P2C	−43.9	+27.2	+8.4
	STPS-SPATIAL	−49.7	+19.8	+8.8
Bursty	RR	−34.1	+22.0	+17.8
	BestFit/DRF	−11.8	+12.5	+19.1
	P2C	−45.2	+25.1	+2.8
	STPS-SPATIAL	−47.3	+15.4	+6.7

Table 8: Cross-regime direction summary for the remaining Stage-B metrics. Phase-shift relieves congestion only where the cap binds (A, C); its CV effect flips sign with binding — it *raises* CV in the hard-binding regime A and lowers it in B and C. p99 tracks congestion: where the cap binds (A, C) draining the queue lowers p99, whereas in the non-binding regime B the offset is pure added delay and p99 rises. The throughput cost is set by the offset budget's ratio to the steady-state window, not by the cap.

Metric direction	A	B	C
CV (non-STPS)	↑	↓	↓
CV (STPS+ps)	↑	↓	↓
JFI	↓	↑	↑
LIF	↑	↓	↓
Max/min	↓	↓	↓
Cong. ratio	↓(.26→0)	—	↓(.09→0)
Throughput cost	18–23%	3–10%	14–16%
p99 (non-STPS)	↓	↑	↓
p99 (STPS+ps)	↓	↑	↓

CV/JFI/LIF directions follow the sign of ΔCV in Table 7.